

8.4 Trigonometric Substitutions

Trigonometric substitutions occur when we replace the variable of integration by a trigonometric function. The most common substitutions are $x = a \tan u$ and $x = a \sin u$.

These substitutions are effective in transforming integrals involving

$$\sqrt{a^2 - x^2} \text{ and } \sqrt{a^2 + x^2}$$

into integrals we can evaluate directly:

With $x = a \tan \theta$,

$$a^2 + x^2 = a^2 + a^2 \tan^2 \theta = a^2(1 + \tan^2 \theta) = a^2 \sec^2 \theta.$$

With $x = a \sin \theta$,

$$a^2 - x^2 = a^2 - a^2 \sin^2 \theta = a^2(1 - \sin^2 \theta) = a^2 \cos^2 \theta.$$

① Substitution $x = a \sin \theta$ for integrals involving $\sqrt{a^2 - x^2}$.

Note 1 We want any substitution we use in an integration to be reversible so that we can change back to the original variable afterward. As we know from Section 7.6, the functions in these substitutions have inverses only for selected values of u . For example,

$$x = a \sin \theta \Leftrightarrow \theta = \arcsin \frac{x}{a}, \left(-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \right).$$

And

$$\sqrt{a^2 - x^2} = \sqrt{a^2(1 - \sin^2 \theta)} = \sqrt{a^2 \cos^2 \theta} = a \cos \theta.$$

Example 1 Evaluate $\int \sqrt{9 - x^2} \, dx$.

Example 2 Evaluate $\int \frac{x^2}{\sqrt{9 - x^2}} \, dx$.

② Substitution $x = a \tan \theta$ for integrals involving $\sqrt{a^2 + x^2}$

Note 2 $x = a \tan \theta \Leftrightarrow \theta = \arctan \frac{x}{a}, \left(-\frac{\pi}{2} < \theta < \frac{\pi}{2} \right).$

And

$$\sqrt{a^2 + x^2} = \sqrt{a^2 (1 + \tan^2 \theta)} = \sqrt{a^2 \sec^2 \theta} = a \sec \theta.$$

Example 3 Evaluate $\int \sqrt{1+x^2} \, dx$.

Example 4 Evaluate $\int \frac{1}{\sqrt{4+x^2}} \, dx$.